American Computer Science League

2023-2024 ● Contest 3: Short Problems Solutions ● Classroom Division

1. Boolean Algebra

$$A + B(\overline{A} + B) = A + \overline{A}B + B$$
$$= A + B(\overline{A} + 1)$$
$$= A + B$$

This is TRUE for (1,0), (0,1) (1,1). So 3 make it TRUE.

D. 3

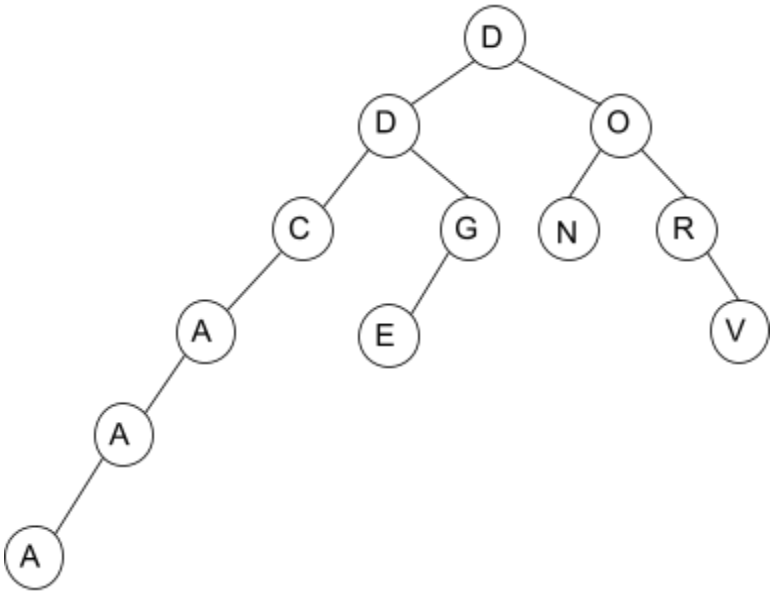
2. Boolean Algebra

$$A + \overline{B + C} + \overline{B} + \overline{\overline{A} + \overline{C}} = A + \overline{B}\,\overline{C} + \overline{B} + \overline{\overline{A}}\,\overline{\overline{C}}$$
$$= A + \overline{B}\,\overline{C} + \overline{B} + A\,C$$
$$= A(1 + C) + \overline{B}\,(\overline{C} + 1)$$
$$= A + \overline{B}$$

C. $A + \overline{B}$

3. Bit-String Flicking

The binary search tree for DODGECARAVAN is:



The depth is 5.

D. 5

4. Data Structures

The stack is constructed using LIFO as follows:
A AL ALP ALPH ALP AL ALA ALAB ALA ALAE
ALAET ALAE
The next item popped would be: E

The queue is constructed using FIFO as follows:
A AL ALP ALPH LPH PH PHA PHAB HAB HABE
HABET ABET
The next item popped would be: A

E. stack top="E",
queue front="A"

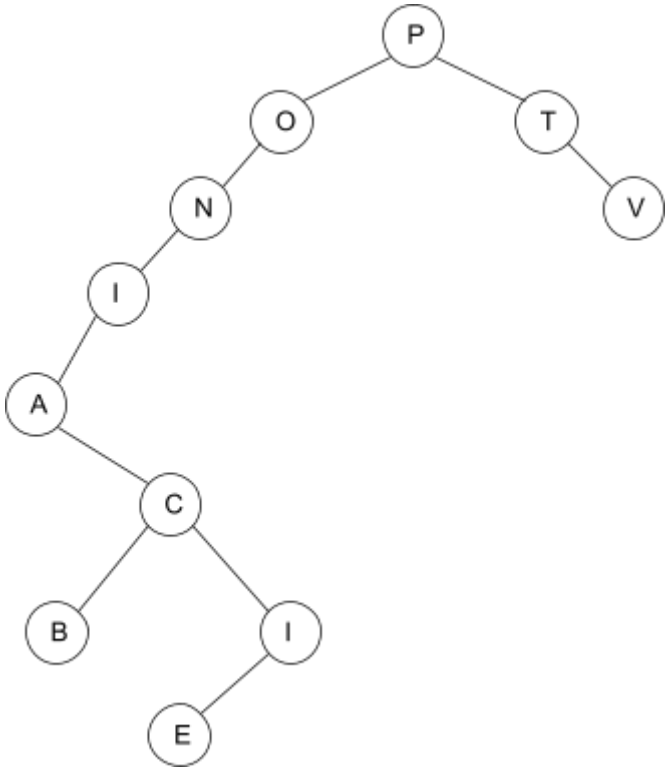
5. Boolean Algebra

$A(\overline{\overline{B} + C})(\overline{AB + BC}) = A\,\overline{\overline{B}}\,\overline{C}\,\overline{AB}\,\overline{BC}$
$= A\,B\,\overline{C}\,(\overline{A} + \overline{B})\,(\overline{B} + \overline{C})$
$= (A\,B\,\overline{C}\,\overline{A} + A\,B\,\overline{C}\,\overline{B})\,(\overline{B} + \overline{C})$
$= (0 + 0)\,(\overline{B} + \overline{C})$
$= 0$

A. 0

6. Data Structures

The binary search tree for PONTIACVIBE is:



The nodes with only one child are: T, O, N, I, A, I. There are 6.

C. 6

7. Data Structures

The stack from top to bottom is A, AL, APL, AHPL, AAHPL, ABAHPL, EABAHPL, ETABAHPL, ESTABAHPL. It puts the letters that are between each vowel in the reverse order of how they appear in the string.

B. ESTABAHPL

8. FSAs

Each of them is parsed as follows:

a. 1101011010 - top path - fails at 5th 1
b. 11010 - top path - fails at 2nd 0
c. 1111111 - fails to reach either path
d. 1010101 - top path - valid
e. 1100110 bottom path - valid
f. 1001110 - bottom path - valid

Therefore, 3 of the strings are valid.

C. 3

9. Regular Expressions

The regular expression: [b-j] [aeiou] [^k]*

Coffees:

regular - failed at r in [b-j]
dark - failed at k in [^k]*
espresso - failed at s in [aeiou]
decaf - valid
drip - failed at r in [aeiou]
black - failed at l in [aeiou]
latte - failed at l in [b-j]
iced - failed at c in [aeiou]
cappuccino - valid

Only 2 match: decaf and cappuccino

A. 2

10. Regular Expressions

The following do not satisfy the given regular expression.

[a-p]* e* [^s]* [d-hp-y]

1. relish fails at h in [d-h, p-y]
2. salt fails at a in [d-h, p-y]
3. mustard fails at t in [d-h,p-y]
4. mayo fails at o in [d-h, p-y]
5. soy fails at o in [d-h, p-y]
6. salsa fails at a in [d-h, p-y]

Therefore, 6 (pickle, pepper, ketchup, butter, vinegar, honey) are valid and 6 fail.

E. 6